

OpenVPN Server Setup on Ubuntu 22.04/24.04

Enable IP Forwarding

Temporary:

```
echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
```

Permanent:

```
sudo vi /etc/sysctl.conf
```

Add:

```
net.ipv4.ip_forward = 1
```

Apply:

```
sudo sysctl -p
```

Verify:

```
cat /proc/sys/net/ipv4/ip_forward
```

Output:

```
1
```

Install OpenVPN and EasyRSA

```
sudo apt update  
  
sudo apt install openvpn easy-rsa wget tar -y
```

Create EasyRSA Working Directory

```
sudo mkdir -p /etc/openvpn/easy-rsa  
  
sudo cp -r /usr/share/easy-rsa/* /etc/openvpn/easy-rsa/  
  
cd /etc/openvpn/easy-rsa
```

Configure EasyRSA

```
sudo vi vars
```

Add:

```
set_var EASYRSA_REQ_COUNTRY "IN"  
set_var EASYRSA_REQ_PROVINCE "Maharashtra"  
set_var EASYRSA_REQ_CITY "Pune"  
set_var EASYRSA_REQ_ORG "Demo Labs"  
set_var EASYRSA_REQ_EMAIL ""  
set_var EASYRSA_REQ_OU "Demo Labs CA"  
set_var EASYRSA_KEY_SIZE 2048  
set_var EASYRSA_ALGO rsa  
set_var EASYRSA_CA_EXPIRE 7500  
set_var EASYRSA_CERT_EXPIRE 365  
set_var EASYRSA_DIGEST "sha256"
```

Save file.

Initialize PKI

```
sudo ./easyrsa init-pki
```

Build Certificate Authority

```
sudo ./easyrsa build-ca nopass
```

Generate Server Certificate

```
sudo ./easyrsa gen-req openvpnserver nopass  
  
sudo ./easyrsa sign-req server openvpnserver  
  
sudo ./easyrsa gen-dh
```

Type:

```
yes
```

when prompted.

Create OpenVPN Directories

```
sudo mkdir -p /etc/openvpn/server  
sudo mkdir -p /etc/openvpn/client
```

Copy Server Certificates

```
sudo cp pki/ca.crt /etc/openvpn/server/  
  
sudo cp pki/dh.pem /etc/openvpn/server/  
  
sudo cp pki/private/openvpnserver.key /etc/openvpn/server/  
  
sudo cp pki/issued/openvpnserver.crt /etc/openvpn/server/
```

Generate Client Certificate

```
sudo ./easysrsa gen-req client1 nopass  
  
sudo ./easysrsa sign-req client client1
```

Type:

```
yes
```

when prompted.

Copy Client Files

```
sudo cp pki/ca.crt /etc/openvpn/client/  
  
sudo cp pki/issued/client1.crt /etc/openvpn/client/  
  
sudo cp pki/private/client1.key /etc/openvpn/client/
```

Create OpenVPN Server Configuration

```
sudo vi /etc/openvpn/server/server.conf
```

Paste:

```
port 1194  
proto udp  
dev tun  
  
ca /etc/openvpn/server/ca.crt  
cert /etc/openvpn/server/openvpnsrvr.crt  
key /etc/openvpn/server/openvpnsrvr.key  
dh /etc/openvpn/server/dh.pem  
  
server 10.8.0.0 255.255.255.0
```

```
push "route 10.0.1.0 255.255.255.0"

duplicate-cn

cipher AES-256-CBC
auth SHA512

tls-version-min 1.2

keepalive 20 60

persist-key
persist-tun

compress lz4

user nobody
group nogroup

verb 3
```

Note:

Replace:

```
push "route 10.0.1.0 255.255.255.0"
```

with your actual LAN network if different.

Start OpenVPN Server

```
sudo systemctl enable openvpn-server@server

sudo systemctl start openvpn-server@server

sudo systemctl status openvpn-server@server
```

Verify:

```
ss -ulnp | grep 1194
```

Expected:

```
udp 0 0 0.0.0.0:1194
```

Create Client Configuration

```
sudo vi /etc/openvpn/client/client1.ovpn
```

Paste:

```
client
dev tun
proto udp

remote 192.168.1.132 1194

ca ca.crt
cert client1.crt
key client1.key

cipher AES-256-CBC
auth SHA512

tls-version-min 1.2

resolv-retry infinite

compress lz4

nobind

persist-key
persist-tun

verb 3
```

Replace:

```
remote 192.168.1.132 1194
```

with your OpenVPN Server NAT IP.

Transfer Client Files

```
scp -r /etc/openvpn/client admin@client-ip:/home/admin
```

Client Machine (Ubuntu)

Install OpenVPN:

```
sudo apt update  
  
sudo apt install openvpn -y
```

Create directory:

```
sudo mkdir -p /etc/openvpn/client
```

Copy files:

```
sudo cp /home/admin/client/ca.crt /etc/openvpn/client/  
  
sudo cp /home/admin/client/client1.crt /etc/openvpn/client/  
  
sudo cp /home/admin/client/client1.key /etc/openvpn/client/  
  
sudo cp /home/admin/client/client1.ovpn /etc/openvpn/client/
```

Connect VPN

```
cd /etc/openvpn/client  
  
sudo openvpn --config client1.ovpn
```

Expected:

```
Initialization Sequence Completed
```

Verification

Open another terminal:

```
ip a
```

Look for:

```
tun0
```

VPN IP:

```
10.8.0.x
```

Check routing:

```
ip route
```

Test:

```
ping 10.0.1.128
```

Ping LAN machine:

```
ping 10.0.1.129
```

If ping works, VPN is functioning correctly.